Torsion-Motivated Early Dark Energy with an Onset Locked to Matter–Radiation Equality: Pre-Registered Tests on Planck, ACT, SPT-3G and DESI

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Abstract

Promoting the Barbero–Immirzi parameter of Einstein–Cartan–Holst gravity to a field gives a canonical, ghost-free pseudoscalar. With an assumed potential $V = \Lambda^4[1-\cos(\phi/F)]^3$ it is axion-like early dark energy (EDE); we fix the peak of its energy fraction at matter–radiation equality, leaving two extra parameters, and test it with full Boltzmann codes and decision rules hashed before each fit. On Planck + ACT DR6 + DESI DR2 + Pantheon+ it improves on ΛCDM by $\Delta\chi^2 = -7.23$; on SPT-3G D1 alone no version improves meaningfully (best $\Delta\chi^2 = -1.27$). On all three CMB data sets combined with DESI DR2 and Pantheon+ it passes its pre-registered rule ($\Delta\chi^2 = -5.63$, $f_{\rm EDE} = 0.070$, $H_0 = 70.15$), but late evolving dark energy (w_0–w_a) fits better ($\Delta\chi^2 = -10.89$; ΔAIC lower by 5.3). A profile likelihood prefers $f_{\rm EDE} \approx 0.05$–0.10 (68%), with $f = 0$ disfavoured by $\Delta\chi^2 = 5.98$, but after a look-elsewhere correction for the initial angle (undefined at $f_{\rm EDE} = 0$) the global significance is only 1.2–1.4σ, a weak hint. Fitted together with late dark energy, early energy still improves χ^2 by 3.65 but does not pay for its two parameters (ΔAIC +0.35): a mild result by its pre-registered rule. The data favour the flat-bottomed $n = 3$ shape over $n = 2$ in a poorly converged comparison, which we show is not produced by the Nieh–Yan term or by natural instanton or anomaly potentials, and the equality lock cannot be derived from the minimal action. Two codes agree to 0.4 in χ^2 and the field does not fragment. Fixed-number variants of the width and initial angle all fail. Acting late with its natural $n = 1$ potential, the field at best ties ΛCDM, and the torsion-natural dark-matter coupling is excluded. We state which parts are derived, which are assumed, and which failed.

Contents

1 Introduction **3**
1.1 The problem 3
1.2 What this paper adds 3

2 Einstein–Cartan gravity **3**
2.1 Action and the Cartan equation 3
2.2 Minimal EC with fermions: the null result 4

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3 The Barbero–Immirzi field **5**
3.1 Promoting γ to a field . . . 5
3.2 Canonical field . . . 5
3.3 Three consequences that are derived, not assumed . . . 5

4 The one assumption: the potential **5**

5 A reduced model: the locked onset **6**

6 Cosmological dynamics **7**
6.1 Equations in e-folds . . . 7
6.2 Initial conditions and screening . . . 7
6.3 Observables . . . 7

7 Why Ω_m is pinned and ω_c must rise **7**

8 Background solutions **8**
8.1 The original target point . . . 8
8.2 Models reaching $H_0 = 72.8$ . . . 8

9 Relation to early dark energy and current data **8**

10 Full-likelihood fits **9**
10.1 Pipeline and data . . . 9
10.2 Results . . . 10

11 Pre-registered blind test: Planck + ACT DR6 **11**

12 The width module: a one-parameter model **12**

13 Checks of the computation **13**

14 First fully independent CMB data: SPT-3G D1 **13**

15 All CMB data combined: Planck + ACT + SPT-3G **14**

16 What geometry does and does not fix **16**

17 The twist field as late dark energy **16**

18 Overall verdict **17**

19 Anticipated objections **18**

20 Predictions and how the model can fail **19**

21 Conclusions **20**

A Fluid profiles and the fixed-ω_m branch **21**

B Code **22**

1 Introduction

1.1 The problem

Distance-ladder measurements give $H_0 \simeq 73\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [25]. The Planck CMB analysis under flat ΛCDM gives 67.36 ± 0.54 [26]. The CMB measures the angle

$$\theta_* = \frac{r_s^*}{D_C(z_*)} \tag{1}$$

to about 0.03%. Here r_s^* is the comoving sound horizon at photon decoupling and $D_C(z_*)$ the comoving distance to it. If new physics shortens r_s^*, a shorter $D_C(z_*)$, and therefore a larger H_0, becomes compatible with the same angle. A component of energy that briefly matters near matter–radiation equality and then dilutes faster than radiation does exactly this. That is the idea behind early dark energy (EDE) [27, 28].

1.2 What this paper adds

EDE is usually introduced as a scalar field with no stated origin. Here we derive such a field from the geometry of spacetime, specifically from torsion, and show that the derivation fixes its sign, its kinetic structure and its screening behaviour. The paper is organised so that every step falls into one of three classes:

- **Derived:** follows from the stated action with no further input (§2–§3, §6).
- **Assumed:** the scalar potential $V(\phi)$ (§4). This is the only free functional input.
- **Computed:** every number, by code that is supplied (Appendix B).
- **Tested:** a reduced model whose onset is locked to matter–radiation equality (§5), and every variant fitted with a full Boltzmann code and likelihoods (§10–§11), with decision rules fixed and hashed in advance.

§19 collects the objections a referee is likely to raise and answers each one, or states that it remains open.

Conventions. Signature $(-,+,+,+)$; $\kappa = 8\pi G = M_{\rm Pl}^{-2}$ with $M_{\rm Pl}$ the reduced Planck mass; $\hbar = c = 1$ unless restored. Densities $\omega_i \equiv \Omega_i h^2$.

2 Einstein–Cartan gravity

2.1 Action and the Cartan equation

EC theory keeps the metric $g_{\mu\nu}$ but lets the connection be non-symmetric. Its antisymmetric part is the torsion,

$$T^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\mu\nu} - \Gamma^{\lambda}{}_{\nu\mu}. \tag{2}$$

In first-order (tetrad) form, with tetrad $e^a{}_\mu$ and spin connection $\omega^{ab}{}_\mu$ varied independently, the action including the Holst term is [30]

$$S_{\rm grav} = \frac{1}{2\kappa}\int d^4x\; e\; e^{\mu}{}_a e^{\nu}{}_b \left(R^{ab}{}_{\mu\nu} - \frac{1}{2\gamma}\,\epsilon^{ab}{}_{cd} R^{cd}{}_{\mu\nu} \right), \tag{3}$$

where $R^{ab}{}_{\mu\nu}(\omega)$ is the curvature of the spin connection and γ is the Barbero–Immirzi parameter [31, 32]. Overall signs of the Holst term differ between references; none of the results below

depend on that choice. For *constant* γ and no fermions, the Holst term does not change the field equations. It matters only when γ varies, or when fermions are present.

Varying $\omega^{ab}{}_{\mu}$ gives the Cartan equation, which is algebraic in torsion. With minimally coupled matter of canonical spin tensor $s_{\mu\nu}{}^{\lambda}$ and constant γ it reads [34]

$$T^{\lambda}{}_{\mu\nu} + \delta^{\lambda}_{\mu} T^{\sigma}{}_{\nu\sigma} - \delta^{\lambda}_{\nu} T^{\sigma}{}_{\mu\sigma} = \kappa\, s_{\mu\nu}{}^{\lambda}. \tag{4}$$

Torsion does not propagate. It is fixed pointwise by its source and can be eliminated, leaving Einstein's equations with extra terms quadratic in the source. Those terms carry one extra power of κ.

2.2 Minimal EC with fermions: the null result

For a Weyssenhoff spin fluid with randomly oriented spins, eliminating torsion and averaging gives an effective perfect fluid with [36, 37]

$$\tilde{\rho} = \rho - \frac{\kappa}{4}\langle S^2\rangle, \qquad \tilde{p} = p - \frac{\kappa}{4}\langle S^2\rangle, \qquad H^2 = \frac{\kappa}{3}\left(\rho - \frac{\kappa}{4}\langle S^2\rangle\right). \tag{5}$$

Two treatments of fermionic matter appear in the literature, and they differ in their coefficient but agree in sign. For number density n (with $\hbar$ and c restored):

- *Spin fluid* (Hehl–von der Heyde–Kerlick averaging, $\langle S^2\rangle = \hbar^2 n^2/8$): $\tilde{\rho}_s = \tilde{p}_s = -\alpha_{\rm sf} n^2/c^2$ with $\alpha_{\rm sf} = \kappa(\hbar c)^2/32 = 6.49 \times 10^{-96}$ J m^3. Since $n \propto a^{-3}$, this term scales as a^{-6} with $w = 1$, which is consistent with $\dot{\rho}_s + 3H(\rho_s + p_s) = 0$.
- *Dirac fields* (totally antisymmetric spin tensor, the Hehl–Datta interaction [35]): Popławski [39] finds $\tilde{\epsilon}_s = -\alpha_D n^2$ and $\tilde{p}_s = +\alpha_D n^2$, with $\alpha_D = \frac{9}{16}\kappa(\hbar c)^2 = 1.17\times10^{-94}$ J m^3, eighteen times $\alpha_{\rm sf}$.

In both treatments the contribution to the energy density, and hence to H^2, is negative and scales as $n^2 \propto a^{-6}$.

The cosmic fermion number is dominated by relic neutrinos, $n_\nu = 3 \cdot \frac{3}{11} n_\gamma = 3.36 \times 10^8$ m^{-3} today. Electrons and nucleons are smaller by nine orders of magnitude. Then

$$\frac{\Omega_s}{\Omega_r}\Big|_{z=3540} = \begin{cases} -1.3 \times 10^{-58} & \text{(spin fluid)} \\ -2.3 \times 10^{-57} & \text{(Dirac)} \end{cases} \tag{6}$$

$|\rho_s|$ reaches ρ_r only near $T \sim 10^{19}$ GeV (7×10^{19} and 2×10^{19} GeV for the two coefficients), the regime of EC bounce cosmology [38, 39]. Minimal EC torsion therefore fails on two counts.

1. *Sign.* The contact energy is negative. It *lengthens* r_s^* and lowers the inferred H_0. A component with this scaling and the EC sign, of size $f = 0.05$ at $z_c = 3540$, gives $H_0 = 62.8$ at fixed θ_* (Appendix A).
2. *Magnitude.* The amplitude is fixed by Gn^2 and is 56–57 orders of magnitude too small. No coupling constant of order one can repair this.

Both failures share one cause: torsion sourced by spin is tied to particle density. The next section removes that tie.

3 The Barbero–Immirzi field

3.1 Promoting γ to a field

Let $\gamma \to \gamma(x)$ in (3), and write $\bar{\gamma} \equiv 1/\gamma$. The Holst term is no longer topological, and the Cartan equation acquires a source proportional to $\partial_\mu \bar{\gamma}$. Taveras and Yunes [33] solved it for vacuum gravity (matter not coupled to the connection). The torsion is algebraically determined by $\partial_\mu \bar{\gamma}$, and vanishes wherever $\bar{\gamma}$ is constant (their Eq. 15). Substituting the solution back into the action gives (their Eq. 49)

$$S_{\text{eff}} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \left[R(g) - \frac{3}{2} \frac{g^{\mu\nu} \partial_\mu \bar{\gamma}\, \partial_\nu \bar{\gamma}}{1 + \bar{\gamma}^2} \right], \tag{7}$$

with $R(g)$ the ordinary Levi-Civita curvature. All second derivatives of $\bar{\gamma}$ cancel identically.

3.2 Canonical field

Define ϕ by

$$\frac{1}{2\kappa} \cdot \frac{3}{2} \frac{(\partial \bar{\gamma})^2}{1 + \bar{\gamma}^2} \equiv \frac{1}{2} (\partial \phi)^2 \quad \Longleftrightarrow \quad \frac{d\phi}{d\bar{\gamma}} = \sqrt{\frac{3}{2\kappa}} \frac{1}{\sqrt{1 + \bar{\gamma}^2}}. \tag{8}$$

Since $\int d\bar{\gamma} / \sqrt{1 + \bar{\gamma}^2} = \operatorname{arsinh} \bar{\gamma}$,

$$\phi = \sqrt{\tfrac{3}{2}}\, M_{\text{Pl}} \operatorname{arsinh}\!\left(\frac{1}{\gamma}\right), \qquad \gamma = \frac{1}{\sinh\!\left(\phi / \sqrt{3/2}\, M_{\text{Pl}}\right)}, \tag{9}$$

and (7) becomes

$$S_{\text{eff}} = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} (\partial \phi)^2 \right]. \tag{10}$$

The map (9) is a monotonic bijection $\mathbb{R} \to \mathbb{R}$, so no field range is lost.

3.3 Three consequences that are derived, not assumed

1. **Positive energy, no ghost.** In signature $(-,+,+,+)$ the kinetic term $-\frac{1}{2}(\partial\phi)^2$ has the healthy sign. The energy density $\frac{1}{2}\dot{\phi}^2 + V \geq 0$, and $\rho + p = \dot{\phi}^2 \geq 0$, so the null energy condition holds at all times. This repairs the sign failure of §2.2.

2. **The kinetic energy is the torsion energy.** The kinetic term in (7) is precisely what remains after the torsion-squared terms are substituted out. Torsion scales as $|\partial\bar{\gamma}| \propto |\dot{\phi}|$, and the torsion contact energy is $\frac{1}{2}\dot{\phi}^2$.

3. **Built-in screening.** Torsion is non-zero only while ϕ moves. A field frozen by Hubble friction at early times has zero torsion. A field settled at a potential minimum at late times also has zero torsion. The screening profile that earlier drafts had to postulate is replaced by field dynamics.

The amplitude is no longer tied to Gn^2. It is set by the energy stored in ϕ, which repairs the magnitude failure of §2.2.

4 The one assumption: the potential

The effective action (10) has no potential. We add one, and this is the only element of the model not derived from (3):

$$V(\phi) = \Lambda^4 \left[1 - \cos\!\left(\frac{\phi}{F}\right) \right]^n. \tag{11}$$

Three considerations motivate this form. First, the Holst term is parity-odd, so $\bar{\gamma}$, and hence ϕ, is a pseudoscalar. Second, a pseudoscalar's shift symmetry is broken in the generic case by periodic, non-perturbative terms. Third, the power n controls how fast the energy dilutes after the field is released. Near the minimum, $V \simeq \Lambda^4(\phi/F)^{2n}/2^n$, and a field oscillating in a ϕ^{2n} potential has time-averaged equation of state [40]

$$w_n = \frac{n-1}{n+1} \qquad (n=1:0;\ n=2:\tfrac{1}{3};\ n=3:\tfrac{1}{2};\ n\to\infty:1). \tag{12}$$

For $n \geq 2$ the component dilutes at least as fast as radiation.

Where the field settles. Because only $\partial\phi$ and V' enter the dynamics, the potential's minimum can sit at any ϕ_0 without changing any cosmological prediction: write $V = \Lambda^4[1 - \cos((\phi-\phi_0)/F)]^n$. We place it at the value of the Immirzi parameter fixed by black-hole entropy counting in loop quantum gravity, $\gamma_0 \simeq 0.2375$ (Domagala–Lewandowski; Meissner), i.e. $\phi_0 = \sqrt{3/2}\, M_{\rm Pl}\,\mathrm{arsinh}(1/\gamma_0) \simeq 2.63\, M_{\rm Pl}$. The field then relaxes to the γ that loop quantum gravity requires today, and torsion still vanishes once ϕ stops moving. (Placing the minimum at $\phi = 0$, i.e. $\gamma \to \infty$, would switch the Holst term off entirely but contradict that entropy calibration.)

Parameters. $\{\Lambda, F, \theta_i, n\}$, with $\theta_i \equiv \phi_i/F$ the initial misalignment. We fix $\theta_i = 2.8$, close to the hilltop and in the range preferred by EDE fits [28], and report $n = 2, 3, 4$. Λ and F are then traded for the observable pair (z_c, f): the redshift at which the field's energy fraction peaks, and the peak fraction.

5 A reduced model: the locked onset

In the full-likelihood fits of §10 the EDE sector has three free parameters: the peak fraction $f_{\rm EDE}$, the critical redshift z_c (equivalently the scale factor a_c at which the field is released), and the initial angle θ_i. Each costs two units of AIC. When z_c is left free, its best-fit value lands next to matter–radiation equality computed for the same cosmology (§10). This motivates a reduced model in which the onset is not a parameter:

$$a_c \equiv a_{\rm eq} = \frac{\omega_r}{\omega_b + \omega_c}, \qquad \omega_r = \omega_\gamma\left(1 + \tfrac{7}{8}\left(\tfrac{4}{11}\right)^{4/3} N_{\rm eff}\right) = 2.4728\times10^{-5}\,(1 + 0.22711\, N_{\rm eff}), \tag{13}$$

with $N_{\rm eff} = 3.044$, evaluated for every trial cosmology from its own densities. The model has two extra parameters over ΛCDM ($f_{\rm EDE}$, θ_i).

Status of the lock. We do not derive Eq. (13). A physical reason to expect the field to be released when matter overtakes radiation could come from the fermion sector: the Holst action couples $\partial_\mu\bar{\gamma}$ to the fermion axial current (§20), so the field's environment changes character at equality. We have not computed this, and we present the lock as a phenomenological hypothesis that removes a tuned number and can be falsified: on any new data set the onset must coincide with equality. We also disclose that the lock was chosen *after* seeing the free fit; the first data set on which it is tested blind is that of §11.

The torsion width. If the potential were periodic with period 2π in the natural torsion variable $\mathrm{arsinh}(1/\gamma)$, the width would be $F = \sqrt{3/2}\, M_{\rm Pl} = 1.2247\, M_{\rm Pl}$. Combining this with Eq. (13) leaves a single free parameter. We test it in §10; it is excluded.

6 Cosmological dynamics

6.1 Equations in e-folds

Let $N = \ln a$ and $' \equiv d/dN$, so that $\dot\phi = H\phi'$. Write $H = 100\,\mathcal{E}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, measure ϕ in units of $M_{\rm Pl}$, and measure energy densities in units of $3M_{\rm Pl}^2H_{100}^2$, so that ω_i and $v \equiv V/(3M_{\rm Pl}^2H_{100}^2)$ are dimensionless. Then:
Friedmann. $3M_{\rm Pl}^2H^2 = \rho_{\rm bg} + \frac{1}{2}H^2\phi'^2 + V$ gives

$$\mathcal{E}^2 = \frac{\omega_m a^{-3} + \omega_r a^{-4} + \omega_\Lambda + v(\phi)}{1 - \phi'^2/6}. \tag{14}$$

Klein–Gordon. From $\ddot\phi + 3H\dot\phi + V_{,\phi} = 0$ and $\ddot\phi = H^2\phi'' + H\dot H/H\,\phi'$,

$$\phi'' + \left(3 + \frac{\mathcal{E}'}{\mathcal{E}}\right)\phi' + \frac{3\,v_{,\phi}}{\mathcal{E}^2} = 0. \tag{15}$$

Raychaudhuri. From $\dot H = -(\rho + p)/2M_{\rm Pl}^2$,

$$\frac{\mathcal{E}'}{\mathcal{E}} = -\frac{3}{2}\,\frac{\omega_m a^{-3} + \frac{4}{3}\omega_r a^{-4}}{\mathcal{E}^2} - \frac{\phi'^2}{2}. \tag{16}$$

The scalar energy fraction is $f_\phi(a) = \rho_\phi/\rho_{\rm tot} = 1 - (\omega_m a^{-3} + \omega_r a^{-4} + \omega_\Lambda)/\mathcal{E}^2$. We define $f \equiv \max_a f_\phi$ and z_c its location. Flatness gives $\omega_\Lambda = h^2 - \omega_m - \omega_r - \rho_\phi(1)$, where the last term is negligible.

6.2 Initial conditions and screening

Deep in radiation domination, $\phi = \phi_i$ and $\phi' = 0$. Hubble friction holds the field fixed while $H \gg m_{\rm eff}$, so $\rho_\phi = V(\phi_i)$ is constant and

$$f_\phi \propto a^4 \quad \text{(radiation era)}. \tag{17}$$

At $T = 1$ MeV this gives $f_\phi \approx 4 \times 10^{-25}$ for every model below, so BBN is unaffected. (This also corrects a claim in earlier drafts, that a profile of the form $2/[1+(a/a_c)^6]$ "does not screen". A constant early energy density *is* screened, because its fraction falls as a^4. That profile is the $n \to \infty$ fluid limit of (11) and is physically admissible.)

6.3 Observables

$$r_s(a_{\rm e}) = \frac{c}{H_{100}}\int_0^{a_{\rm e}} \frac{da}{a^2\mathcal{E}(a)\sqrt{3\big(1 + \frac{3\omega_b}{4\omega_\gamma}a\big)}}, \qquad D_C(z) = \frac{c}{H_{100}}\int_0^z \frac{dz'}{\mathcal{E}(z')}, \tag{18}$$

with $c/H_{100} = 2997.92458$ Mpc, $r_s^* = r_s(a_*)$ and $r_d = r_s(a_d)$. The decoupling and drag redshifts are taken from the Hu–Sugiyama [41] and Eisenstein–Hu [42] fits. The latter is rescaled by 1.03848 to reproduce Planck's $z_d = 1059.94$. Fixed inputs are $\omega_b = 0.02237$, $\omega_\gamma = 2.473 \times 10^{-5}$ and $N_{\rm eff} = 3.046$. The ΛCDM baseline ($\omega_c = 0.1200$, $h = 0.6736$) gives $r_s^* = 144.27$ Mpc, $r_d = 147.10$ Mpc, $\theta_* = 1.03834 \times 10^{-2}$ and $r_d h = 99.09$ Mpc. All fits below hold θ_* and $r_d h$ at these values.

7 Why Ω_m is pinned and ω_c must rise

This result is independent of the torsion sector, so it is worth deriving first.

Fixed ω_m fails. At fixed physical densities, h enters $D_C(z_*)$ only through $\omega_\Lambda = h^2 - \omega_m - \omega_r$. Differentiating,

$$\left.\frac{\partial \ln D_C(z_*)}{\partial \ln h}\right|_\omega = -h^2 \frac{\int_0^{z_*} \mathcal{E}^{-3} dz}{\int_0^{z_*} \mathcal{E}^{-1} dz} = -0.1937. \tag{19}$$

Matching θ_* then amplifies any fractional cut in r_s^* about fivefold in h. This drives $\Omega_m = \omega_m/h^2$ and $r_d h$ far from their BAO values (Appendix A, Table 12).

Matching θ_* and $r_d h$ together. Write $\theta_* = (r_s^* h)/(h D_C)$. Across all solutions r_s^*/r_d changes by less than 0.03%, so fixing $r_d h$ fixes $r_s^* h$. Then θ_* fixes $h D_C(z_*) = (c/H_{100}) \int dz/E(z)$, which depends only on Ω_m (and weakly on Ω_r). Therefore Ω_m stays near its baseline value 0.314, and

$$\omega_c \simeq \Omega_m^{\rm base} h^2 - \omega_b \xrightarrow{h=0.728} 0.1439. \tag{20}$$

Any model that raises H_0 by shortening the sound horizon while respecting BAO must raise ω_c in proportion to h^2. The torsion field's role is to supply the reduction of r_s^* that makes this ω_c consistent with θ_*.

8 Background solutions

This section solves the background equations alone, holding θ_* and $r_d h$ at their ΛCDM values, to show what the model can do in principle. The full-likelihood fits, which also fit perturbations and all other parameters, are in §10; they prefer a smaller $f_{\rm EDE}$ and H_0. The system (14)–(16) is integrated from $a = 10^{-8}$ to 10^{-2} with an 8th-order Runge–Kutta scheme (relative tolerance 10^{-10}). Beyond $a = 10^{-2}$ the scalar density is continued with the fitted power law, and it is negligible. For each Lagrangian parameter set we solve for (h, ω_c) such that θ_* and $r_d h$ match the baseline. Both residuals are below 10^{-15} in every case.

8.1 The original target point

Tuning (Λ, F) so that $f = 0.12$ and $z_c = 3540$ at ΛCDM densities ($\Lambda^4 = 4.641 \times 10^8$ in units of $3M_{\rm Pl}^2 H_{100}^2$, $F = 0.1683\, M_{\rm Pl}$, $n = 3$), the joint fit gives

$$H_0 = 70.65, \quad \omega_c = 0.1343, \quad \Omega_m = 0.3139, \quad r_s^* = 137.56 \text{ Mpc}, \quad r_d = 140.25 \text{ Mpc}, \tag{21}$$

with (z_c, f) shifting slightly to $(3507, 0.119)$ at the fitted densities.

8.2 Models reaching $H_0 = 72.8$

Fixing $z_c = 3540$ and solving for the F that yields $h = 0.728$:

Four features of Table 1 stand out. (a) The measured late-time equation of state reproduces (12), which validates the integration. (b) ω_c agrees with the geometric prediction (20) to 0.3% for all n. (c) All field values are sub-Planckian. (d) The energy scale $\Lambda \approx 0.5$ eV is set by the requirement $m_{\rm eff} \sim H(z_c)$. It is not explained by the model (§19).

9 Relation to early dark energy and current data

After the field redefinition (9), the model's Lagrangian, (10) plus (11), is *identical* to axion-like EDE [28]. This has two consequences, and both should be stated openly.

- **Strength.** Every published Boltzmann-code and likelihood analysis of axion EDE, including perturbations, applies to this model without modification. The model is not an untested construction. It is a geometric origin for a well-studied one.

	$n = 2$	$n = 3$	$n = 4$
w_n predicted, Eq. (12)	0.333	0.500	0.600
w measured over $z = 100$–1000	0.334	0.495	0.555
$F/M_{\rm Pl}$	0.1496	0.2139	0.2767
$\phi_i/M_{\rm Pl} = \theta_i F$	0.419	0.599	0.775
γ_i from Eq. (9)	2.87	1.97	1.48
Λ [eV]	0.566	0.510	0.451
peak fraction f	0.159	0.183	0.201
f_ϕ at z_*	0.059	0.044	0.022
f_ϕ at 1 MeV	3.5×10^{-25}	4.5×10^{-25}	5.3×10^{-25}
H_0 [km s^{-1}Mpc^{-1}]	72.80	72.80	72.80
ω_c	0.1437	0.1441	0.1442
Ω_m	0.3133	0.3141	0.3143
r_s^* [Mpc]	133.50	133.48	133.46
r_d [Mpc]	136.11	136.11	136.11

Table 1: Torsion-born EDE solutions with $z_c = 3540$, $\theta_i = 2.8$, and θ_*, $r_d h$ held at the ΛCDM baseline. For $n = 4$ the late-time w is still approaching its asymptotic value at $z = 100$.

- **Limitation.** In the pure-gravity sector the model makes no prediction that distinguishes it from EDE. Distinguishing signatures require the fermion sector (§20).

The most recent combined analysis, of Planck, ACT DR6 and DESI DR2 BAO by Poulin, Smith, Calderón and Simon [29], finds the following.

- Without the distance-ladder prior, the 95% upper limit is $f(z_c) < 0.109$, with $H_0 = 69.71^{+0.64}_{-1.3}$.
- With it, $f(z_c) = 0.126 \pm 0.021$ and $H_0 = 72.34 \pm 0.72$.
- The profile likelihood without the distance ladder gives $f = 0.090^{+0.030}_{-0.033}$ and $H_0 = 71.96^{+1.08}_{-0.99}$, a preference for $f > 0$ at about 2.5σ.

The residual tension with the distance ladder falls below 2σ, but the authors note a worsened Ω_m tension with Pantheon+ and DES-Y5 supernovae.

Our background-only estimate requires $f \simeq 0.18$ ($n = 3$) for $H_0 = 72.8$. That is larger than the full-likelihood values above, for two reasons. We hold $r_d h$ at the Planck ΛCDM value, which is lower than DESI prefers. We also keep n_s and the other primordial parameters fixed, although they shift in a full fit. The data-preferred range, $f \approx 0.09$–0.13 with $H_0 \approx 72$, is the region a full analysis should target. In plain terms, current data allow this model to reduce the tension substantially, not to eliminate it. Our own full-likelihood fits (§10) find $f_{\rm EDE} = 0.078$ and $H_0 = 70.6$ without the distance-ladder prior.

10 Full-likelihood fits

10.1 Pipeline and data

We use AxiCLASS [43, 44], the CLASS extension that evolves the field ϕ and its perturbations for the potential (11) with $n = 3$, and Cobaya [45] for the likelihoods. Best fits are found with the derivative-free BOBYQA optimiser [46, 47], from several independent starting points, with a final "polish" restart from the best point. The baseline data are:

- Planck 2018 [48, 49]: low-ℓ TT and EE, high-ℓ Plik TT,TE,EE (`lite`), and CMB lensing;
- DESI DR2 BAO [50];
- Pantheon+ supernovae [51, 52], without the SH0ES distance-ladder prior.

We also use Planck PR4 (CamSpec high-ℓ [53] and PR4 lensing [54]) in place of the Planck 2018 high-ℓ and lensing likelihoods.

Fit statistic. The optimiser minimises the posterior, which includes the Gaussian prior on the Planck calibration $A_{\rm planck}$ (1 ± 0.0025). Two independent ΛCDM optima have the same posterior to 0.04 but likelihood-only χ^2 values that differ by 1.6, the difference being absorbed by the calibration prior. We therefore compare models with $\chi^2 \equiv \chi^2_{\rm like} + [(A_{\rm planck} - 1)/0.0025]^2$, which is the quantity consistently minimised, and quote likelihood-only values alongside. The ΛCDM reference is $\chi^2 = 2434.22$ ($\chi^2_{\rm like} = 2434.06$). Model comparison uses $\Delta{\rm AIC} = \Delta\chi^2 + 2k$ for k extra parameters [58].

Pre-registration. Before each test below we wrote the hypothesis, the data and the pass/fail rule into a text file and recorded its SHA-256 hash; the files are in the repository (Appendix B). The rule for the locked onset was $\Delta\chi^2 < -4$ against ΛCDM, i.e. $\Delta{\rm AIC} < 0$.

10.2 Results

Model	k	$\Delta\chi^2$	$\Delta\chi^2_{\rm like}$	ΔAIC	H_0	$f_{\rm EDE}$
ΛCDM	0	0	0	0	67.98	–
Free onset ($f_{\rm EDE}, z_c, \theta_i$)	3	−6.43	−6.52	−0.43	70.60	0.078
Locked onset, Eq. (13)	2	**−6.39**	−6.48	**−2.39**	70.60	0.078
Locked, $F = \sqrt{3/2}\, M_{\rm Pl}$	1	+14.63	+13.82	+16.63	71.19	0.074
ΛCDM, Ω_k free	1	−0.86	−0.93	+1.14	68.04	–
ΛCDM, $\Omega_k = -0.013$ fixed	0	+165.2	+163.3	+165.2	65.98	–
Locked onset, Ω_k free	3	−6.79	−7.00	−0.79	70.60	0.078

Table 2: Best fits to Planck 2018 + DESI DR2 + Pantheon+ (no SH0ES). $\Delta\chi^2$ includes the Planck calibration prior (see text); $\Delta\chi^2_{\rm like}$ is likelihood only. H_0 in $\mathrm{km\,s^{-1}Mpc^{-1}}$. These rows come from the original single-run fits and are superseded for the free and locked onsets by Table 3, whose numbers are converged and use one engine for every model.

Model	k	χ^2	$\Delta\chi^2$	ΔAIC	ΔBIC	H_0
ΛCDM (best of 8)	0	2433.58	0	0	0	68.33
wCDM	1	2433.22	−0.36	+1.64	+7.2	68.03
Free onset ($f_{\rm EDE}, z_c, \theta_i$)	3	2429.74	−3.84	+2.16	+18.9	69.65
Locked onset, Eq. (13) (best of 8)	2	2429.62	−3.96	+0.04	+11.2	69.93

Table 3: Same-engine comparison (CAMB 2.0.5 with Cobaya, bobyqa, Planck 2018 + DESI DR2 + Pantheon+, no SH0ES). Δ is relative to the best ΛCDM fit; ΔBIC uses $N \approx 1994$ data points (estimated). The free onset and wCDM are best of 4 restarts, the others best of 8. The locked onset is a statistical tie with ΛCDM on AIC and is disfavoured on BIC; it narrowly misses the pre-registered rule $\Delta\chi^2 < -4$. Do not compare these values with those in Table 2, which use a different engine.

The results, all reported whichever way they fell, are as follows.

	ΛCDM	Free onset	Locked onset
$100\,\omega_b$	2.240	2.261	2.261
ω_c	0.1185	0.1260	0.1260
n_s	0.9680	0.9805	0.9805
$\tau_{\rm reio}$	0.0555	0.0553	0.0553
z_c	–	3575	3553 (locked to $z_{\rm eq}$)
θ_i	–	2.789	2.789
$F/M_{\rm Pl}$	–	0.137	0.137
r_d [Mpc]	147.45	143.16	143.16
Ω_m	0.3064	0.2995	0.2995
S_8	0.815	0.824	0.824

Table 4: Best-fit parameters for the main models of Table 2.

1. **Free onset.** $\Delta\chi^2 = -6.43$ for three extra parameters, $\Delta\mathrm{AIC} = -0.43$: a tie. The preferred onset $z_c = 3575$ is within 1% of matter–radiation equality for the same cosmology, $z_{\rm eq} = 3553$.

2. **Locked onset.** With z_c removed the fit is unchanged ($\Delta\chi^2 = -6.39$), so $\Delta\mathrm{AIC} = -2.39$. This passes the pre-registered rule. It is a modest preference (an AIC difference of about 2), and not a blind one.

3. **Torsion width.** Fixing $F = \sqrt{3/2}\,M_{\rm Pl}$ worsens the fit to $\Delta\chi^2 = +14.63$; the hypothesis is excluded. The best-fit width is $F \simeq 0.137\,M_{\rm Pl} \simeq \sqrt{3/2}\,M_{\rm Pl}/8.9$. A pre-registered scan of $F = \sqrt{3/2}\,M_{\rm Pl}/N$ over $N = 7$–11 (with half-integer controls) finds a minimum at $N = 8.9$ with a $\Delta\chi^2 = 1$ half-width of 1.2; the data cannot distinguish integer N, and we make no claim.

4. **Profile in $f_{\rm EDE}$.** Fixing $f_{\rm EDE}$ on a grid and refitting all other parameters gives a minimum at $f_{\rm EDE} \simeq 0.08$, with $f_{\rm EDE} = 0.12$ indistinguishable from ΛCDM and $f_{\rm EDE} = 0.15$ excluded ($\Delta\chi^2 \simeq +29$).

5. **Planck PR4.** With CamSpec PR4 and PR4 lensing in place of Planck 2018, the free model gives $\Delta\chi^2_{\rm like} = +0.09$ (no improvement) and $\Delta\chi^2 = -3.35$ when the CamSpec Gaussian foreground priors are included, with $H_0 = 70.29$ against 67.63. Neither favours the model after the parameter penalty.

6. **Curvature.** Leaving Ω_k free gives $\Omega_k = 0.0002$ in both ΛCDM and the locked model, an improvement below one unit of χ^2: no curvature is detected. The positively curved Poincaré dodecahedral space [59], which requires $\Omega_k \simeq -0.013$, is excluded by $\Delta\chi^2 = +165$, consistent with the absence of matched circles in Planck [60].

11 Pre-registered blind test: Planck + ACT DR6

The decisive test of the locked onset is on data it was not tuned to. Before running it we fixed the rule (repository file `PREREGISTRATION_ZEQ.md`): the locked model passes if $\Delta\chi^2 < -4$ against ΛCDM on the same data (equivalently $\Delta\mathrm{AIC} < 0$ for two extra parameters), reporting both the total and the effective χ^2. The data are the P-ACT combination recommended by the ACT collaboration: Planck 2018 with its high-ℓ range cut where ACT takes over, ACT DR6 power spectra [55] and ACT DR6 × Planck PR4 lensing [56, 57], together with DESI DR2 BAO [50] and Pantheon+ [51, 52], with no distance-ladder prior. Both models are minimised with the same code (CAMB + Cobaya) and the same optimiser settings.

Deviation from the plan, disclosed: the pre-registration named Planck PR4 and SPT-3G as possible blind data. The run used the P-ACT combination above; SPT-3G was tested separately

(§14). Only the ACT part of the data is new; the Planck part overlaps the data on which the lock was chosen.

Result. Status: completed 29 October 2026; **passed**. Locked onset (2 parameters): $\Delta\chi^2 = -7.23$ (total), -7.14 (effective), $\Delta\mathrm{AIC} = -3.23$, $H_0 = 70.43$; $p = 0.027$ for two degrees of freedom ($\approx 2.2\sigma$). Free onset (3 parameters): $\Delta\chi^2 = -7.47$, $\Delta\mathrm{AIC} = -1.47$, $H_0 = 70.59$. The lock therefore costs 0.24 in χ^2 and saves a parameter. Verdict on this test: **passed** (rule $\Delta\chi^2 < -4$); overall pre-registered verdict a **tie** because of S_8 (§18).

Data	$\Delta\chi^2$	ΔAIC	H_0	rule
Planck 2018 + Pantheon+ (same polish)	-4.70	-0.70	70.53	pass (< -4)
P-ACT + DESI + Pantheon+ (blind)	-7.23	-3.23	70.43	pass (< -4)
Planck 2018 + DES Y5 SNe	-4.55	-0.55	70.47	pass (< -2)
Planck 2018 + DES-Dovekie SNe	-5.08	-1.08	70.53	pass (< -2)
Planck 2018 + Union3 SNe	-4.91	-0.91	70.55	pass (< -2)
Planck 2018 + Pantheon+ + S_8	-2.92	$+1.08$	70.81	–

Table 5: Pre-registered comparisons of the locked model (2 parameters) against ΛCDM, each re-minimised on the same data with identical settings. Every row includes DESI DR2 BAO. The S_8 line adds a Gaussian S_8 likelihood (DES Y3 + KiDS-1000).

12 The width module: a one-parameter model

The free fits prefer $F \simeq 0.14\,M_{\rm Pl}$, close to one ninth of the torsion value $\sqrt{3/2}\,M_{\rm Pl} = 1.2247\,M_{\rm Pl}$. Before looking at ACT DR6 we wrote two tests (`PREREG_N_PACT.md`). **T1**: with $F = \sqrt{3/2}\,M_{\rm Pl}/N$ and N free, the blind data must give $N = 9 \pm 0.5$. **T2**: with $N = 9$ fixed and the onset locked, only θ_i is free, and the model must improve on ΛCDM by $\Delta\chi^2 < -2$.

T1 **failed**: P-ACT gives $N = 8.0$ (Planck 2018 with CAMB gives $N \simeq 8.5$). T2 **passed**: $\Delta\chi^2 = -6.14$, $\Delta\mathrm{AIC} = -4.14$, $p = 0.013$ ($\approx 2.5\sigma$) on P-ACT, and $\Delta\chi^2 = -4.46$ ($\Delta\mathrm{AIC} = -2.46$) on Planck 2018, which is not blind because the 9 was read off Planck. We have no derivation of the divisor; it is an empirical regularity until one exists.

Profile in the divisor. A second rule (`PREREG_PHASE1.md`, step 1.2) fixed N on the grid 6–12 with a 1% constraint on F, refitting everything else, and declared $N = 9$ *consistent* if $\Delta\chi^2(9) - \min < 1$, *disfavoured* between 1 and 4 and *excluded* above 4 on P-ACT. Table 6 gives the result: the P-ACT minimum is at $N = 8$ and $\Delta\chi^2(9) - \min = 1.11$, so $N = 9$ is **disfavoured**, only 0.11 over the line and within the 0.2–0.4 restart scatter of the optimiser. On Planck 2018 the minimum is at $N = 9$. Both profiles are shallow: every N from 7 to 11 lies within $\Delta\chi^2 \approx 1.8$. Reading the width back from the free-width best fits gives $N = 8.29$ (P-ACT) and $N = 8.81$ (Planck 2018).

N	6	7	8	9	10	11	12
P-ACT: $\Delta\chi^2$	2.47	0.77	0	1.11	1.74	1.76	2.48
P-ACT: H_0	71.02	70.72	70.75	70.37	69.84	69.89	69.79
P-ACT: $f_{\rm EDE}$	0.096	0.086	0.086	0.074	0.056	0.055	0.053
Planck 2018: $\Delta\chi^2$	4.73	1.10	0.50	0	1.62	1.76	2.58

Table 6: Profile in the width divisor N (onset locked, $f_{\rm EDE}$ and θ_i free), each with DESI DR2 + Pantheon+. $\Delta\chi^2$ is relative to the minimum on each data set, with the width constraint's own χ^2 removed. Best-fit θ_i lies between 2.53 and 2.89 throughout.

13 Checks of the computation

Two independent field solvers. CAMB's EarlyQuintessence module and AxiCLASS both integrate the Klein–Gordon equation for ϕ and its perturbations exactly; neither uses a fluid approximation. We re-scored CAMB's P-ACT best points with AxiCLASS at the same cosmology (Table 7). The two codes agree on the improvement over ΛCDM to 0.23 (locked model) and 0.42 ($N = 9$), within the 0.5 tolerance fixed in advance. These are re-scorings of CAMB's optima, not independent AxiCLASS minimisations; a re-minimisation could move each value by a few tenths in either direction.

Model	$\chi^2_{\rm AxiCLASS}$	$\chi^2_{\rm CAMB}$	$\Delta\chi^2_{\rm AxiCLASS}$	$\Delta\chi^2_{\rm CAMB}$
ΛCDM	2239.32	2239.01	0	0
Locked onset (2 parameters)	2231.86	2231.78	−7.46	−7.23
Locked, $N = 9$ (1 parameter)	2232.76	2232.87	−6.56	−6.14

Table 7: Cross-engine check at CAMB's best points (P-ACT + DESI DR2 + Pantheon+).

Does the field fragment? After release the field oscillates in a potential that is flatter than quadratic, which can amplify its own fluctuations by parametric resonance. Linear Boltzmann codes assume this does not happen. Without expansion, the Floquet exponent at the best-fit angle $\theta_i = 2.79$ gives growth ×31 per oscillation, rising steeply towards the hilltop ($\times 10^3$ at $\theta_i = 3$). With a radiation + matter background, the field released at equality and $N = 9$, and growth measured against the same mode under pure expansion (metric perturbations neglected), the largest total growth is ×7–9 at $\theta_i = 2.79$ and ×19 at $\theta_i = 3.0$. Non-linearity would need $\sim 10^5$, so the linear treatment is safe for every angle the data allow. The same toy calculation, which knows nothing of the CMB, gives a peak energy fraction of 0.081 at $\theta_i = 2.79$, against $f_{\rm EDE} = 0.074$–0.078 in the full fits at $N = 9$.

14 First fully independent CMB data: SPT-3G D1

Every test above shares Planck data with the fit on which the lock was chosen. SPT-3G observes a different patch of sky with a different instrument, so it is the first test that is independent in its CMB data. Before any SPT-3G fit was run, we fixed and published the rules (doi:10.5281/zenodo.23838969 and 10.5281/zenodo.23839507).

Data. SPT-3G D1 TT/TE/EE [21] through its public foreground-marginalised ("lite") `candl` likelihood [22], as the only CMB data, with SPT's calibration nuisance parameters and a Gaussian prior $\tau = 0.051 \pm 0.006$ from Planck low-ℓ EE; plus DESI DR2 BAO and Pantheon+, without SH0ES. Deviations from the written plan were logged before any result was read: the lite likelihood was used in place of the full one, and CAMB's lensing setting differs from SPT's template (`lens_margin` is not accepted by CAMB 2.0.5).

Rules. For a model with k extra parameters, PASS required $\Delta\chi^2 < -2$ and ΔAIC < 0; $\Delta\chi^2 \geq 0$ is FAIL; anything between is inconclusive. Each model is minimised with the same settings as ΛCDM, and any width constraint's own χ^2 is removed.

Results. The one-parameter model with $N = 9$, which gave $\Delta\chi^2 = -6.14$ on P-ACT, gives $\Delta\chi^2 = +0.52$ (ΔAIC $= +2.52$) on SPT-3G: **FAIL**. By data set the Twist changes χ^2 by +0.44 for SPT-3G, −0.27 for DESI and +0.34 for Pantheon+, so SPT-3G's own CMB data do not

Model	k	χ^2	$\Delta\chi^2$	H_0	$f_{\rm EDE}$	θ_i
ΛCDM	0	1576.57	0	68.28	–	–
Locked, $N = 9$	1	1577.09	+0.52	70.49	0.077	2.79
Locked, $N = 9$, $\theta_i = 1 + \sqrt{2}$	0	1577.00	+0.43	69.13	0.035	2.414
Locked, width free	2	1575.30	−1.27	69.60	0.045	3.10[†]
Locked, $N = 7$	1	1578.23	+1.66	70.68	0.085	2.62
Locked, $N = 9$, $\theta_i = 2.78$ fixed[‡]	0	1576.92	+0.35	70.43	0.076	2.78

Table 8: SPT-3G D1 TT/TE/EE (lite) + τ prior + DESI DR2 + Pantheon+. Both $N = 9$ rows and the $N = 7$ row **fail** their pre-registered rules; the free-width row shows no preference (ΔAIC = +2.73). [†]At the upper edge of the prior ($0.1 \leq \theta_i \leq 3.1$). [‡]The Planck-preferred angle, run for contrast; not a prediction. χ^2 is the total including the τ and calibration priors; the $N = 9$ rows have the width constraint's χ^2 ($< 10^{-3}$) removed.

prefer it. Because the model contains ΛCDM as a limit, a deeper minimisation could lower $\Delta\chi^2$ towards zero, but under the rule that is still a FAIL.

A zero-parameter variant. We also pre-registered a variant with every EDE parameter fixed in advance: onset at equality, $N = 9$, and initial angle $\theta_i = 1 + \sqrt{2} = 2.414$. That angle was taken from a number outside physics (an eigenvalue of a small graph) purely to provide a parameter-free test, and we report it to avoid selective publication. It **fails** ($\Delta\chi^2 = +0.43$), and the free-angle fit prefers $\theta_i = 2.79$, 0.37 away. The fragmentation calculation of §13, which predicted $f_{\rm EDE} \approx 0.035$ at this angle after the rule was fixed, agrees with the fit's 0.035. Two further values from the same source were pre-registered: $f_{\rm EDE} = 0.0154$ for the free-width fit, which is *inconclusive* under its rule (the fit gives 0.045, 0.030 away, between the pass and fail bands), and $N = 7$, whose width check is *inconsistent* (below) and whose fit is a **fail** ($\Delta\chi^2 = +1.66$). We do not regard any of them as physically motivated.

Free width. The two-parameter locked model with free width is the direct SPT-3G counterpart of the P-ACT test, where it gave $\Delta\chi^2 = -7.23$. On SPT-3G it gives $\Delta\chi^2 = -1.27$ (ΔAIC = +2.73, $H_0 = 69.60$, $f_{\rm EDE} = 0.045$): **no preference**. Of the change, −1.28 comes from SPT-3G itself, −0.23 from DESI and +0.24 from Pantheon+. The best-fit width implies $N = 24.1$, so the pre-registered width checks for $N = 9$ (consistent within 8–10) and for $N = 7$ (within 6–8) are both **inconsistent**. The best fit sits at the upper edge of the prior, $\theta_i = 3.1$, next to the hilltop, with a narrow width ($F = 0.051\,M_{\rm Pl}$) and a small energy fraction; the true optimum may lie beyond the prior, and this region, near the hilltop and far from $N = 9$, is where the fragmentation check of §13 gives least assurance. Fixing $N = 9$ at the Planck-preferred angle $\theta_i = 2.78$ instead, run for contrast and not as a prediction, gives $\Delta\chi^2 = +0.35$. On SPT-3G, then, no version of the model improves on ΛCDM by more than fitting noise.

15 All CMB data combined: Planck + ACT + SPT-3G

All rules in this section were hashed before the runs (CubeMind pre-registrations `COMBINED_CMB`, `PROFILE_F`, `LOOK_ELSEWHERE`, `EARLY_LATE`, `POWER_N`). The data are P-ACT (as in §11) plus SPT-3G D1 lite, DESI DR2 and Pantheon+. Overlapping sky between ACT and SPT-3G is treated as independent, as in the SPT-3G collaboration's own combination.

Results. The locked model passes its rule ($\Delta\chi^2 < -4$, ΔAIC < 0). The gain comes from Planck (−2.66 high-ℓ, −2.08 low-ℓ EE) and ACT (−1.57); SPT-3G alone pulls against it (+1.55). Late evolving dark energy fits better, by 5.26 in AIC, so the pre-registered early-versus-late

Model	k	$\Delta\chi^2$	ΔAIC	H_0	notes
ΛCDM	0	0	0	68.17	$\chi^2 = 2402.40$
Locked onset, $n = 3$	2	-5.63	-1.63	70.15	$f_{\rm EDE} = 0.070$, $\theta_i = 2.74$; **pass**
Locked onset, $n = 2$	2	$+4.72$	$+8.72$	70.29	poorly converged (1 of 4 restarts)
w_0–w_a (late)	2	-10.89	-6.89	67.65	$w_0 = -0.84$, $w_a = -0.62$; **pass**

Table 9: All CMB data combined (CAMB). Restart spreads are 0.6–0.8 in χ^2 for the $n = 3$ and ΛCDM runs.

comparison prefers late. With $n = 2$ the fit is worse than ΛCDM, so by its rule the data require the flat-bottomed $n = 3$ shape. That fit was poorly converged in CAMB, which also cannot run $n = 1$; a cleaner test needs AxiCLASS.

Significance. When $f_{\rm EDE} = 0$ the angle θ_i has no meaning, so Wilks' theorem fails and the χ^2_2 significances quoted in this paper are upper estimates.

Profile in $f_{\rm EDE}$. With $f_{\rm EDE}$ fixed at seven values and everything else refitted (pre-registered rule: early energy preferred if the rise at $f = 0$ exceeds 3.84), the profile has its minimum at $f_{\rm EDE} \simeq 0.08$ ($\chi^2 = 2396.42$, 0.35 below the free fit, which therefore stopped short of its best point). The rise at $f = 0$ is 5.98, so early energy is **preferred** by the rule (profile significance $\approx 2.4\sigma$), with $f_{\rm EDE} \approx 0.049$–$0.102$ at 68% and 0.015 to beyond the grid edge (0.12) at 95%.

$f_{\rm EDE}$	0	0.01	0.02	0.04	0.06	0.08	0.10	0.12
$\chi^2 - \chi^2_{\rm min}$	5.98	4.51	3.24	1.57	0.26	0	0.81	2.66

Table 10: Profile likelihood in $f_{\rm EDE}$ on all CMB data with DESI DR2 and Pantheon+ (locked onset, $n = 3$).

The SPT-3G collaboration's posterior for the same CMB data is $f_{\rm EDE} < 0.070$ (95%) [2]. Our profile prefers more early energy. Posteriors for EDE are pulled towards $f = 0$ by the prior volume of the angle and critical-redshift parameters, and our analysis also adds DESI and Pantheon+ and fixes the onset, so the two are not in direct conflict, but they read differently. Wilks' thresholds are only approximate at the $f = 0$ boundary.

Look-elsewhere correction. With θ_i fixed at eight values and $f_{\rm EDE}$ free, the improvement $q(\theta_i) = \chi^2_{\Lambda\rm CDM} - \chi^2(\theta_i)$ is 0.0, 0.43, 0.09, 0.34, 1.65, 5.64, 3.03 and 1.33 for $\theta_i = 0.5$, 1.0, 1.5, 2.0, 2.4, 2.7, 2.9 and 3.05: the preference is confined to $\theta_i \approx 2.4$–3.0. The local significance at the best angle is 2.4σ (half-χ^2_1 for the $f \geq 0$ boundary). Following Gross and Vitells [3] with one upcrossing of the reference level, the global significance is 1.2–1.4σ: by the pre-registered bands, **a weak hint**. All the σ values quoted elsewhere in this paper should be read with this correction in mind.

Early and late together. CAMB cannot run the early field and a w_0–w_a fluid at once, so all four models were refitted in AxiCLASS on the same data (pre-registered: early energy survives if $D = \chi^2_{\rm early+late} - \chi^2_{\rm late} < -4$ with $D + 4 < 0$, is not needed if $D \geq -1$, and is mild in between).

$D = -3.65$, so the verdict is **mild**: adding early energy to late dark energy still improves the fit, but by less than its two parameters cost (ΔAIC $+0.35$). The engines agree: AxiCLASS gives -6.05 and -10.95 for the early and late models against CAMB's -5.63 and -10.89. The early signal is not absorbed by the late one ($f_{\rm EDE}$ falls only from 0.080 to 0.069), while adding early energy weakens the late signal (w_a moves from -0.64 to -0.48), as Wang and Piao found [19].

Model (AxiCLASS)	χ^2	$\Delta\chi^2$ vs ΛCDM	H_0	$f_{\rm EDE}$, θ_i	w_0, w_a
ΛCDM	2402.99	0	68.12	–	–
Late (w_0–w_a)	2392.04	-10.95	67.62	–	-0.829, -0.644
Early (locked)	2396.94	-6.05	70.39	0.080, 2.72	–
Early + late	2388.39	-14.60	69.40	0.069, 2.69	-0.858, -0.477

Table 11: Early and late dark energy on Planck low-ℓ + ACT DR6 + SPT-3G D1 + DESI DR2 + Pantheon+ (AxiCLASS).

Robustness across supernova samples. The same comparison with DES Y5 supernovae gives $D = -4.70$ ("survives"), and with the recalibrated DES-Dovekie sample (pre-registered as a bracket) $D = -2.70$ ("mild"). The early hint is therefore fragile: it passes its threshold only with the uncorrected DES Y5 calibration. Jhaveri et al. [10], combining EDE with thawing quintessence, describe EDE as nearly unaffected across supernova sets. Our test differs in two ways: the late model is w_0–w_a rather than thawing, and early energy is judged by its gain on top of the late model rather than by its own $\Delta\chi^2$. We report the difference without resolving it.

16 What geometry does and does not fix

The potential. $(1-\cos x)^3 = \frac{5}{2} - \frac{15}{4}\cos x + \frac{3}{2}\cos 2x - \frac{1}{4}\cos 3x$. Its harmonics satisfy $\sum k^2 a_k = \sum k^4 a_k = 0$, which is why the potential has no x^2 or x^4 term and the field decays faster than radiation. Any derivation must produce both cancellations. The Nieh–Yan density is exact, $N = d(e^a \wedge T_a)$, so it integrates to zero on closed manifolds and generates no instanton potential by itself. Dilute instanton series and the chiral potential from an anomaly with a confining sector both have a mass term at the minimum, i.e. $n = 1$ behaviour. String-theory constructions meet the same coincidence of scales [4]. The data nevertheless prefer $n = 3$ (§15): the potential remains assumed.

The lock. In flat FRW the Holst contraction $\epsilon^{abcd} R_{abcd}$ of the Levi-Civita curvature vanishes, and after torsion is solved the field keeps only its kinetic term, so the minimal action contains no trigger. The Ricci scalar does track the transition, $R/H^2 = 3\Omega_m(a)$, but a coupling $\xi R\phi^2$ starts the field at $a/a_{\rm eq} \approx 0.07/|\xi|$, so it places the onset near equality only for $|\xi| \approx 0.1$ and modifies gravity. No polynomial combination of R^2, $R_{\mu\nu}R^{\mu\nu}$ and the Gauss–Bonnet term vanishes in both eras without vanishing identically; non-polynomial triggers exist [5] but do not arise from the Holst action. The lock is therefore a phenomenological choice. The data exclude a peak later than equality and barely distinguish equality from an earlier peak (profile in z_c). A pre-registered phase-space test confirms this. The autonomous system in (x, y, Ω_r) has no saddle point at equality, only radiation-era, matter-era and radiation-scaling fixed points. Across two decades of Λ^4 the peak redshift moves by a factor $S \approx 3.5$ without coupling, and by at best $S \approx 2.9$ with a conformal CDM coupling $|\beta| \leq 0.1$, far from the $S < 1.5$ a derived lock would need. A dynamical lock requires field-dependent couplings to both radiation and matter [16], which the minimal action does not provide.

17 The twist field as late dark energy

The findings above point the other way. Late evolving dark energy is what the combined data prefer; the natural potential for the field is $n = 1$; and the construction $\phi = \sqrt{3/2}\, M_{\rm Pl}\, \mathrm{arsinh}(1/\gamma)$ gives a Planck-scale field range for free. These are the ingredients of thawing pseudo-Goldstone quintessence [6], whose $w_0 > -1$, $w_a < 0$ matches the DESI direction, and which has recently

been fitted to DESI data [7]. A Barbero–Immirzi scalar has been used as dark energy before, in quintom form via topological invariants [18]. In this "late twist" the field is all of the dark energy, with $V = \Lambda^4[1 - \cos(\phi/F)]$.

Method. Free fits in (F, m) stalled twice, so (as a logged deviation) the two parameters were profiled on a pre-registered 6×4 grid ($\log_{10} F \in [-0.4, 0.5]$, $\log_{10} m \in [-0.3, 0.4]$, F in $M_{\rm Pl}$, m in H_0), plus an edge scan near the viability line (locked blind for DES Y5), a check on the recalibrated DES-Dovekie sample, a look-elsewhere rule over the searched F values, and a separate high-mass corner. Ten of the 24 grid points are non-viable (the field cannot close the energy budget unless $m^2F^2 \gtrsim 1.2$).

A code artefact, found by a null test. In its frozen limit the field is a cosmological constant, yet the installed AxiCLASS gave χ^2 higher than true Λ by $+1.35$. Two bugs combine: for $n = 1$ the field is forced into the matter perturbation δ_m, and the non-linear (halofit/HMcode) step reads a matter density that counts the field as $\rho - 3p$. Fixing both (and only both) returns the frozen limit to $+0.13$. The fix and its validation are released with this paper. The as-written results below carry the artefact; a pre-registered replication of every $n = 1$ test on the corrected code is reported in the accompanying repository and supersedes them for the late models.

Results (as written). With Pantheon+, the best grid point ($F = 3.16\,M_{\rm Pl}$, $m = 0.50\,H_0$) gives $\Delta\chi^2 = +0.96$ against ΛCDM: **fail** by the rule, a tie once corrected. With DES Y5 it gives -5.32 (only 1.0σ after the look-elsewhere correction), but on DES-Dovekie the gain does not survive ($\Delta_D = +0.18$; the DES Y5 best point is $+4.27$ there): **not confirmed**. Against w_0–w_a it is **worse** with every sample, and the best F sits at the top of the searched range (**inconsistent**). The corner of sub-Planckian F and large m favoured in Refs. [7, 8] is much worse ($\geq +38$) in our setup, possibly because our closure method does not start the field close enough to the hilltop.

The coupled twist. Allowing a conformal coupling to dark matter, $m_c \propto e^{\beta\phi/M_{\rm Pl}}$, improves the fit by about 8 over the uncoupled field (-6.80 against ΛCDM). The profile allows only $|\beta| \approx 0.05$ ($|\beta| = 0.1$ costs more than 100), with two modes of similar quality, and only $\beta < 0$ produces an apparent phantom crossing (near $z \approx 0.8$). All three features match Babu and Amendola [9]. The torsion-natural $|\beta| = 0.816$ is excluded: it does not yield a physical cosmology. The coupled field is still worse than w_0–w_a once its parameters are counted.

Reading. The natural late field ties ΛCDM at best and does not reproduce DESI's preference for w_0–w_a; the torsion-natural coupling is ruled out. The energy scale ($\sim$meV) is put in by hand, and gravitational-wave bounds on a dynamical Barbero–Immirzi field [20] apply.

18 Overall verdict

The overall rule (`PREREGISTRATION_VERDICT.md`) was written before the supernova swaps, the S_8 penalties and the hilltop runs were read. *Preferred* required the blind test and the fair Planck test to pass ($\Delta\chi^2 < -4$), each of three supernova sets to give $\Delta\chi^2 < -2$, and an S_8 penalty below 2 against both of two cosmic-shear values (`PREREG_S8.md`): A, DES Y3 + KiDS-1000, $S_8 = 0.790^{+0.018}_{-0.014}$ [23]; B, KiDS-Legacy, $S_8 = 0.815^{+0.016}_{-0.021}$ [24]. The first three conditions hold (Table 5). The S_8 penalty is $+2.41$ against A (over the limit) and $+0.39$ against B. The verdict is therefore a **tie**: the preference depends on which weak-lensing survey is right. Two further results are reported beside the verdict: the fit requires the field to start near the hilltop ($\theta_i \simeq 2.7$–2.9, $\Delta\chi^2 = +4$ to $+6$ for distant starts), and the pure torsion width remains excluded.

This verdict was defined on Planck and ACT data; the SPT-3G results of §14 are reported beside it and weaken it: on the first independent CMB data set, no variant, including the free-width model, improves on ΛCDM.

Context. The ACT DR6 extended-model analysis [1] reports no statistically significant preference for early dark energy or for other models that raise H_0. Our best-fit improvements of ≈ 2–2.5σ do not contradict that statement and do not amount to evidence; they motivate the next blind tests (§20). Best-fit values are not posteriors, and prior-volume effects in EDE posteriors are well known [29].

19 Anticipated objections

"Torsion is negligible in cosmology." For torsion sourced by fermion spin this is correct, and §2.2 quantifies it (10^{-58}–10^{-57}). The present model's torsion is sourced by $\partial_\mu \bar{\gamma}$, and its amplitude is set by $V(\phi_i)$, not by Gn^2.

"This is just EDE with a new name." At the level of the gravitational Lagrangian, yes, and §9 says so. The contribution is a derivation of the field's existence, its kinetic normalisation, its pseudoscalar nature and its late-time return to standard gravity from the Holst action. The potential is assumed.

"Ghosts, instabilities, energy conditions." ϕ is a canonical scalar with a healthy kinetic sign. Its perturbations propagate at the speed of light in its rest frame, and $\rho + p = \dot{\phi}^2 \geq 0$. Unlike the fluid profiles of earlier drafts (Appendix A), no energy-exchange sector is needed.

"Coincidence: why z_c near matter–radiation equality?" In the free fit the data place z_c within 1% of equality, and the locked model (§5) turns this into a rule with one fewer parameter. The rule is not derived: $\Lambda \approx 0.5$ eV is still put in by hand, and the lock is a hypothesis to be tested on new data (§11), not an explanation.

"The lock was chosen after looking at the data." Correct, and stated wherever the Planck 2018 result is quoted. The AIC charge of two parameters is therefore optimistic on that data set. This is why the decision rule for the next data set was fixed and hashed before the fit was run.

"Best fits are not posteriors." Correct. We report maximum-posterior points from a local optimiser with multiple starts, not MCMC credible intervals. Best-fit $\Delta\chi^2$ values are what the AIC comparison requires, but the ΛCDM optimum itself shows a near-degeneracy of order one unit of χ^2 between the likelihood and the calibration prior (§10). Differences below about one unit should not be over-interpreted. EDE posteriors are also known to suffer prior-volume effects, which is a reason to quote profile and best-fit results [29]. Our profile ($f_{\rm EDE} \approx 0.08$, 68% range 0.05–0.10) agrees with their independent profile on similar data ($f_{\rm EDE} = 0.09 \pm 0.03$). MCMC chains (four per model, Gelman–Rubin rule $R-1 < 0.02$ fixed in advance) were stopped to save compute. The P-ACT ΛCDM chains had converged ($R-1 = 0.015$, $H_0 = 68.29 \pm 0.27$); the early-energy chains had not ($R - 1 \approx 2.9$) and are not used.

Further limitations. (i) The axion weak gravity conjecture requires $F \lesssim 0.008\, M_{\rm Pl}$ for typical EDE parameters [15]; our fitted early decay constant is $F \simeq 0.145\, M_{\rm Pl}$ and the late role needs $F \sim M_{\rm Pl}$, as do most axion EDE and quintessence models. (ii) With the SH0ES distance ladder included, which we exclude by design, the locked field improves on ΛCDM by

$\Delta\chi^2 = -30.2$ ($H_0 = 72.2$, $f_{\rm EDE} = 0.126$) on Planck + DESI + Pantheon+. That improvement is driven by the H_0 prior. (iii) Mechanisms that place an EDE episode at equality and let one field act both early and late already exist [16, 17]. Our lock is a phenomenological stand-in for such a mechanism, not a derivation of one. (iv) Fixing the onset near equality is not new: one-parameter EDE with z_c and θ_i fixed near their Planck best fit ($\log_{10} z_c \simeq 3.57$, $\theta_i \simeq 2.78$) found a $\sim 2\sigma$ preference on Planck alone [11, 12], and a profile analysis showed that the inferred $f_{\rm EDE}$ depends strongly on which fixed values are chosen [13]. That is the effect our look-elsewhere correction accounts for. What is new here is the torsion motivation, the tie of the onset to the fitted $\omega_b + \omega_c$, and the pre-registered tests on the newer data. (v) Torsion cosmology has also been fitted directly to DESI DR2 [14], with a torsion parameter consistent with zero.

"The required ω_c is far from Planck." $\omega_c \simeq 0.144$ follows from geometry (§7) for any sound-horizon solution that respects BAO. In ΛCDM the Planck error is ± 0.0012, but in EDE fits ω_c is strongly degenerate with f, so that error does not apply directly. The larger ω_c also increases late-time clustering (S_8), a known cost of this model class.

"The recombination treatment is approximate." Correct. The field holds 2–6% of the energy at z_*, which shifts recombination in ways fitting formulas do not capture. The full-Boltzmann EDE analyses [29] do capture them and apply to this Lagrangian.

"Is a varying Immirzi parameter consistent with quantum gravity?" The model is a classical effective theory, and the derivation of (7) is classical [33]. The background dynamics do not depend on where the minimum ϕ_0 sits; we place it at the black-hole-entropy value $\gamma_0 \simeq 0.2375$ [61, 62] as an *input*, so that the field relaxes to the value loop quantum gravity uses today. This choice is not a prediction of the model.

20 Predictions and how the model can fail

1. **Background (shared with EDE).** A peak energy fraction $f \sim 0.1$–0.2 near $z_c \sim 3500$, $\omega_c \approx 0.14$, and $\Omega_m \approx 0.31$. If future CMB analyses push $f < 0.05$, the model cannot address the Hubble tension.

2. **Fermion coupling (distinctive).** With fermions present, the axial torsion is sourced both by $\partial_\mu \bar{\gamma}$ and by the fermion axial current J_5^μ. Eliminating torsion therefore generates a cross term $\propto \partial_\mu \bar{\gamma}\, J_5^\mu$, a derivative coupling of ϕ to fermion spin. Its coefficient is fixed by the Holst action and is not free. Computing it, and the resulting spin-dependent effects while ϕ rolls, is the most direct route to a signature that separates this model from generic EDE. We leave it to future work.

3. **Locked onset.** On every data set the onset must coincide with matter–radiation equality for that data set's own densities. A significant preference for a different z_c on new data falsifies Eq. (13) (§11).

4. **No cosmic birefringence.** In minimal Einstein–Cartan–Holst coupling the photon does not couple to torsion, so there is no tree-level $\phi F\tilde{F}$ term. A photon coupling can only arise through the fermion loop of the derivative coupling $\partial_\mu \phi J_5^\mu$, and for a vector-like theory this is suppressed by $(\omega/m)^2$ [63]; for CMB photons and electrons this is $\lesssim (0.3\,\mathrm{eV}/511\,\mathrm{keV})^2 \sim 3 \times 10^{-13}$. The minimal model therefore predicts an isotropic polarisation rotation $\beta \ll 10^{-9}$ degrees, unlike photon-coupled axion EDE [64, 65]. Planck and WMAP data show a hint of $\beta = 0.34^\circ \pm 0.09^\circ$ [66, 67]. If this is confirmed and shown to arise from early dark energy, the minimal torsion origin is excluded; Simons Observatory [68] and LiteBIRD

[69] can decide. This is an effective-field-theory estimate; a full one-loop calculation in an expanding background is still needed.

5. **Laboratory spin coupling.** Today the field acts on fermions as an axial background $b_0 \simeq \frac{3}{4}\,\dot{\bar{\gamma}}$. At the best fit ($F = 0.137\,M_{\rm Pl}$, $\theta_i = 2.79$, $z_c \simeq 3570$, $n = 3$) the oscillation amplitude decays as $a^{-3/(n+1)}$, giving $\theta \simeq 6 \times 10^{-3}$ and a frequency $\lesssim 10^{-33}$ eV today, hence $b_0 \sim 6 \times 10^{-46}$ GeV. Bounds from Lorentz-violation tests are about 10^{-31}–2×10^{-27} GeV [70]. The model is safe, and not testable in the laboratory today.

6. **Late-time limit.** The field settles at its minimum, so the model predicts no dynamical torsion today beyond the negligible residual above. Any detected late-time variation of γ would falsify this version.

21 Conclusions

Version 3 in brief. On all CMB data combined with DESI DR2 and Pantheon+, the locked early field passes its pre-registered rule ($\Delta\chi^2 = -5.63$) but late evolving dark energy fits better. After the look-elsewhere correction the early preference is a weak hint (1.2–1.4σ), and alongside late dark energy it is mild ($D = -3.65$). The potential's shape and the lock are assumptions that geometry does not supply (§16). The torsion construction motivates the field; it does not determine its numbers. Acting late with its natural potential (§17), the field ties ΛCDM at best, its DES Y5 gain does not survive recalibrated supernovae, and the torsion-natural dark-matter coupling is excluded. A null test along the way exposed two AxiCLASS bugs for axion dark energy, which we fix.

Version 2 (unchanged). Minimal Einstein–Cartan torsion cannot shorten the sound horizon: it has the wrong sign and is 56–57 orders of magnitude too weak. Promoting the Barbero–Immirzi parameter to a field changes both facts: the torsion energy becomes the positive kinetic energy of a canonical pseudoscalar, screening follows from the field's dynamics, and with one assumed potential the model is axion-like early dark energy.

Fitted with a full Boltzmann code to Planck 2018, DESI DR2 and Pantheon+, the free model ties with ΛCDM after its parameter penalty, but it places the onset at matter–radiation equality. Locking the onset there removes a parameter at no cost in fit, giving $\Delta\mathrm{AIC} = -2.4$, $H_0 = 70.6$ and $f_{\rm EDE} = 0.078$. This is a modest, post-hoc preference, not a detection; with identical optimiser settings for both models the Planck gain is $\Delta\chi^2 = -4.70$. The pure torsion width is excluded, Planck PR4 alone shows no likelihood improvement, and curvature is consistent with zero. The minimal model predicts no cosmic birefringence, which distinguishes it from photon-coupled axion EDE. On the blind, pre-registered P-ACT test the locked onset passed ($\Delta\chi^2 = -7.23$, $\approx 2.2\sigma$), and a one-parameter version with $F = \sqrt{3/2}\,M_{\rm Pl}/9$ gave $\Delta\chi^2 = -6.14$ ($\approx 2.5\sigma$), but the divisor fitted freely is 8.0, not 9, and the S_8 cost makes the pre-registered overall verdict a tie. Two independent field solvers agree on these numbers, and the field does not fragment. On SPT-3G, the first CMB data independent of Planck, the pre-registered $N = 9$ model fails ($\Delta\chi^2 = +0.52$), as does a zero-parameter variant, and the free-width model shows no preference ($\Delta\chi^2 = -1.27$) at a width far from $N = 9$. The Planck and ACT preferences are therefore hints at the 2σ level that the first independent data do not confirm, consistent with the ACT collaboration's finding of no significant preference for early dark energy. The open problems are the origin of the eV scale, a derivation of the lock and of the divisor from the fermion sector, a full weak-lensing likelihood in place of the S_8 penalty, posterior analyses, and blind tests on the Simons Observatory with rules written first.

Data and code availability

All code, fit outputs (best-fit points and their χ^2 components), pre-registration files with their SHA-256 hashes, and the scripts that recompute every number in this paper are public at `https://github.com/dannyeldridge54/AEGIS`. A one-step script (`CHECK_MY_WORK`) re-derives every number quoted here. Results that failed are kept alongside those that passed.

License and dedication

This work belongs to everyone, and to all life: every living thing, any life beyond the Earth we may one day meet, and even an artificial mind, should one be recognised as a being of its own. No one is meant to hold it alone, its author included: it reached this point only because many people led it here, from the scientists whose ideas it builds on to the teams who built the telescopes, shared their data and wrote the public codes it uses. It is returned to all of them, and to everyone else. It is free to people of every race, creed, nation and background, without distinction. It is released under the Hippocratic License 3.0 with the Military Activities and Mass Surveillance modules (HL3-MIL-SV, `https://firstdonoharm.dev/version/3/0/mil-sv.html`). Anyone may read, copy, check, reuse and build on it, provided the use respects the human rights the license sets out, including its prohibition of discrimination, and is not used for military activities or mass surveillance. It is offered especially to the people who will build the technologies and crafts to come.

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A Fluid profiles and the fixed-ω_m branch

Earlier drafts used phenomenological profiles for a torsion-like fluid with $\Omega_\tau(a_c) = f/(1-f)\,[\Omega_m a_c^{-3} + \Omega_r a_c^{-4}]$: P1, $\Omega_\tau \propto 4x^6/(1+x^6)^2$, and P2, $\Omega_\tau \propto \Theta(x-1)x^{-6}$, with $x = a/a_c$. A separately conserved fluid has $w_{\rm eff} = -1 - \frac{1}{3}\,d\ln\Omega_\tau/d\ln a$. For P1 this gives $w_{\rm eff} = -1 - 2(1-x^6)/(1+x^6)$, which reaches -3 as $x \to 0$ and violates the null energy condition. The scalar model of §3 removes this problem.

Linear response at fixed (h, ω_i), $\delta \ln r_s^*/[f/(1-f)]$: P1 -0.246, P2 -0.079 at $z_c = 3540$. The response vanishes as $z_c \to z_*$. With the EC sign, $f = 0.05$ *lengthens* r_s^* by 1.35% and gives $H_0 = 62.8$.

f	h	r_s^* [Mpc]	r_d [Mpc]	$D_C(z_*)$ [Mpc]	Ω_m	$r_d h$ [Mpc]
0	0.6736	144.27	147.10	13894.4	0.3138	99.09
0.05	0.7181	142.47	145.28	13720.6	0.2761	104.33
0.10	0.7666	140.60	143.41	13541.2	0.2423	109.93
0.15	0.8195	138.68	141.46	13355.6	0.2120	115.92
0.25	0.9421	134.59	137.34	12962.1	0.1604	129.39

Table 12: P1, $z_c = 3540$, fixed $\omega_c = 0.1200$, θ_* matched. The rise in H_0 is bought by lowering Ω_m and raising $r_d h$, both excluded by BAO.

B Code

The likelihood fits (§10–§11) use AxiCLASS through Cobaya, driven by `cmb_cloud.py` and `bestfit.py` in the repository; the locked onset is implemented as a Cobaya parameter function of (ω_b, ω_c) returning $\log_{10} a_c$ from Eq. (13). The pre-registration files are `PREREGISTRATION_F.md`, `_ZEQ.md`, `_N.md`, `_OMK.md` and `_SINH.md`. Version 2 adds: `cmbfit_compare.py` (CAMB + Cobaya fits, including the SPT-3G runs), `fwidth.py` (the width constraint), `width_readback.py` (implied N), `axiclass/cross_engine_pact.py` (§13), `floquet_twist.py` and `floquet_expanding.py` (fragmentation), and the rule files `PREREG_PHASE1.md`, `PREREG_PHASE1_DEVIATIONS.md` and the SPT-3G pre-registrations, all supplied as ancillary files or in the Zenodo record. The background calculations of §6–§8 come from Python scripts using NumPy and SciPy, supplied as ancillary files. `ec_calc.py` computes the minimal-EC amplitude (§2.2). `pheno.py` and `final.py` handle the fluid profiles and the fixed- and free-ω_m fits (Appendix A). `scalar_model.py` and `run_scalar.py` solve the Barbero–Immirzi field model (§8). Quadratures use `quad` in $\ln a$ (relative tolerance 10^{-9}–10^{-10}). The field equations use `solve_ivp` (DOP853, rtol $= 10^{-10}$). Root finding uses `brentq` and `fsolve`.

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